

1. An object oscillates with simple harmonic motion along the x axis. Its displacement from the origin varies with time according to the equation $x = 4.00 \cos(\pi t + \pi/4)$ where t is in seconds and the angles in the parentheses are in radians.
 - a) Determine the amplitude, frequency and period of the motion.
 - b) Calculate the velocity and acceleration of the object at any time t.
 - c) Using the results in part (b), determine the position, velocity and acceleration of the object at $t = 1.00$ s.
 - d) Determine the maximum speed and maximum acceleration of the object.
2. A block with a mass of 200 g is connected to a light spring for which the force constant is 5.00 N/m and is free to oscillate on a horizontal, frictionless surface. The block is displaced 5.00 cm from equilibrium and released from rest.
 - a) Find the period of this motion.
 - b) Determine the maximum speed of the block.
 - c) What is the maximum acceleration of the block?
 - d) Express the displacement, speed and acceleration as functions of time.
3. An object is undergoing SHM with period 0.900 s and amplitude 0.320 m. At $t = 0$, the object is at $x = 0.320$ m and is instantaneously at rest. Calculate the time it takes the object to go (a) from $x = 0.320$ m to $x = 0.160$ m and (b) from $x = 0.160$ m to $x = 0$.
4. A 0.500 kg mass on a spring has velocity as a function of time given by $v_x(t) = -(3.60 \text{ cm/s}) \sin[(4.71 \text{ s}^{-1})t - \pi/2]$. What are (a) the period; (b) the amplitude; (c) the maximum acceleration of the mass; (d) the force constant of the spring?
5. A mass m at the end of a spring oscillates with a frequency of 0.83 Hz. When an additional 680 g mass is added to m, the frequency is 0.60 Hz. What is the value of m?
6. A 175-g glider on a horizontal, frictionless air track is attached to a fixed ideal spring with force constant 155 N/m. At the instant you make measurements on the glider, it is moving at 0.815 m/s and is 3.00 cm from its equilibrium point. Use energy conservation to find (a) the amplitude of the motion, and (b) the maximum speed of the glider. (c) What is the angular frequency of the oscillations?
7. A bullet is fired into a 4.648 kg wooden block resting on a smooth surface, and attached to a spring of spring constant $k = 142.7$ N/m. The bullet, whose mass is 7.870 g, remains embedded in the wooden block. The maximum distance that the block compresses the spring is 9.46 cm. What is the speed v of the bullet?
8. A block of mass 1.6 kg is attached to a horizontal spring that has a spring constant of 1.0×10^3 N/m. The spring is compressed 2.0 cm and then released from rest. Calculate the speed of the block as it passes through the equilibrium position $x = 0$
 - (a) if the surface is smooth.
 - (b) if a constant frictional force of 4.0 N retards its motion from the moment it is released.

9. A simple pendulum is made of a string of length 0.240 m and mass 0.10 kg. You pull it to the side at an angle of 3.50 degrees to the vertical and release it. (a) How much time does it take the pendulum bob to reach its highest speed? (b) How much time does it take if the pendulum is released at an angle of 1.75 degrees instead of 3.50 degrees?
10. A simple pendulum oscillates with amplitude of 10.0° . Its time period is T. If initially the pendulum was at its maximum displacement, what fraction of time (in terms of its time period T) does it spend between $+5.0^\circ$ and -5.0° during one oscillation cycle? Assume simple harmonic motion.

Answers

- 1 a) $A = 4.00$ m, $f = 0.500$ Hz, $T = 2.00$ s b) $v = -4.00\pi \sin(\pi t + \pi/4)$, $a = -4.00\pi^2 \cos(\pi t + \pi/4)$
c) $x = -2.83$ m, $v = 8.89$ m/s, $a = 27.9$ m/s² d) $v_{\max} = 12.6$ m/s, $a_{\max} = 39.5$ m/s²
2. a) $T = 1.26$ s b) $v_{\max} = 0.250$ m/s
c) 1.25 m/s² d) $x = 0.05 \cos 5.00t$, $v = -0.250 \sin 5.00t$, $a = -1.25 \cos 5.00t$
3. a) 0.150 s b) 0.0750 s
4. a) $T = 1.33$ s b) $A = 7.64$ mm c) $a_{\max} = 0.169$ m/s² d) 11.1 N/m
5. $m = 0.74$ kg
6. a) $A = 4.06$ cm b) $v_{\max} = 1.21$ m/s c) $\omega = 29.8$ rad/s
7. $v = 309.8$ m/s
8. a) $v = 0.50$ m/s b) $v = 0.39$ m/s
9. a) $t = 0.25$ s b) $t = 0.25$ s
10. $t = T/3$